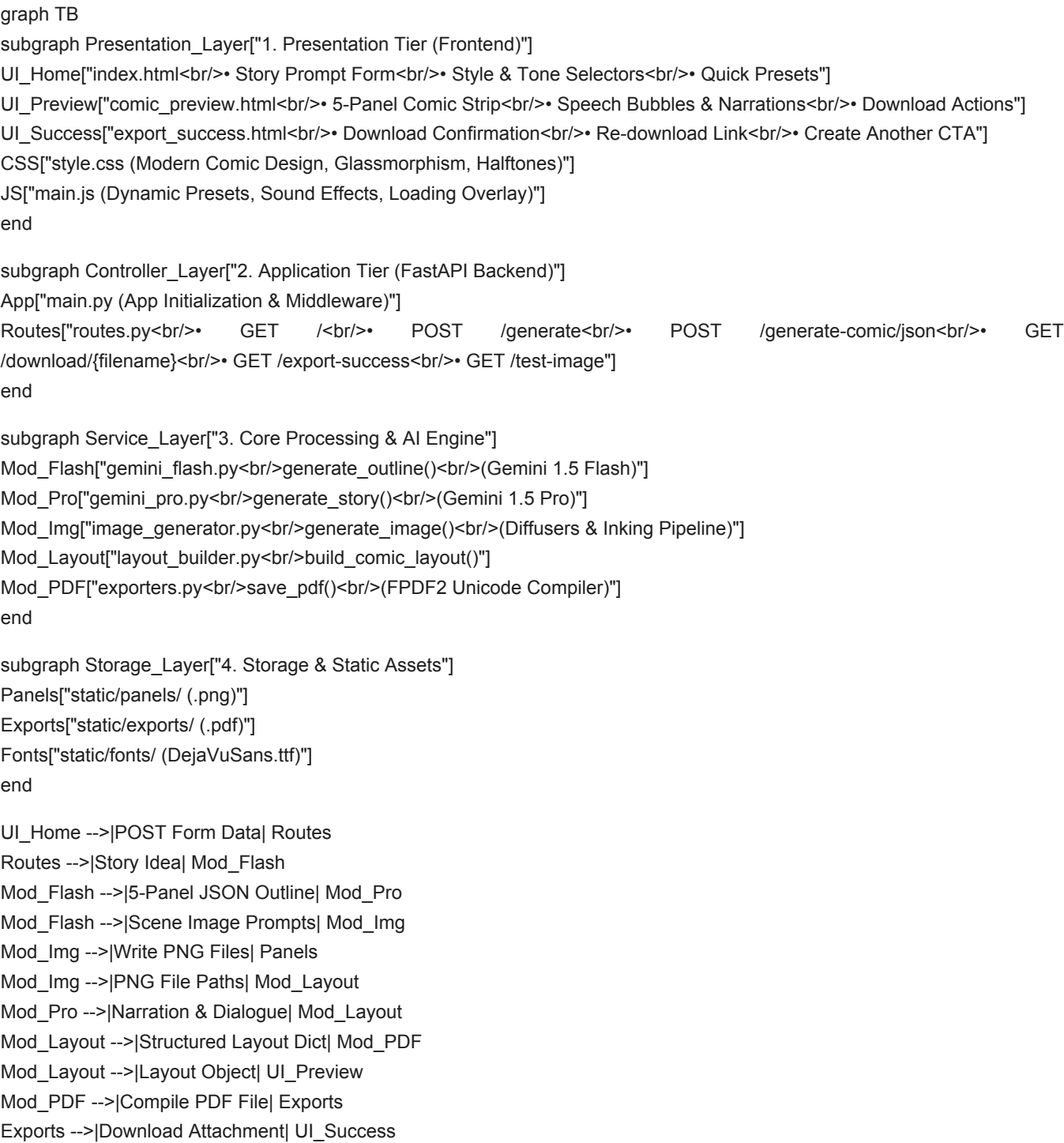


Phase 3: Project Design Phase

Solution Architecture

1. Architectural Diagram

ComicCraft follows a clean, three-tier modular architecture separating presentation, business logic orchestration, and AI model services:



2. Component Responsibilities

Component	File Path	Primary Responsibility
UI_Home	index.html	Story Prompt Form
UI_Preview	comic_preview.html	5-Panel Comic Strip
UI_Success	export_success.html	Download Confirmation
CSS	style.css	Modern Comic Design, Glassmorphism, Halftones
JS	main.js	Dynamic Presets, Sound Effects, Loading Overlay
App	main.py	App Initialization & Middleware
Routes	routes.py	API Endpoints
Mod_Flash	gemini_flash.py	generate_outline()
Mod_Pro	gemini_pro.py	generate_story()
Mod_Img	image_generator.py	generate_image()
Mod_Layout	layout_builder.py	build_comic_layout()
Mod_PDF	exporters.py	save_pdf()
Panels	static/panels/	Comic Panels (.png)
Exports	static/exports/	Comic Exports (.pdf)
Fonts	static/fonts/	Comic Fonts (DejaVuSans.ttf)

| Server Entrypoint | app/main.py | Instantiates FastAPI, configures CORS middleware, mounts /static file directories, and registers route controllers. |

| API Router | app/routes.py | Handles incoming HTTP requests, performs input validation via Pydantic, coordinates the AI pipeline, and renders templates. |

| Panel Outliner | app/gemini_flash.py | Queries Gemini 1.5 Flash with structured system instructions to produce a valid 5-panel JSON storyline array. |

| Dialogue Expander | app/gemini_pro.py | Queries Gemini 1.5 Pro to expand the outline into comic book narration, environmental captions, and character dialogue. |

| Image Generator | app/image_generator.py | Inks panel scenes using generative diffusion or artistic illustration pipelines, outputting optimized PNG assets. |

| Layout Builder | app/layout_builder.py | Extracts title, speech bubble, caption, and narration metadata, mapping them to corresponding panel imagery. |

| PDF Exporter | app/exporters.py | Formats each panel into a dedicated A4 page with TrueType Unicode fonts and compiles the downloadable PDF document. |

3. Fault-Tolerance & Resilience Strategy

- Decoupled Fallback Engine: If the Gemini API is unreachable or environment keys are absent, both gemini_flash.py and gemini_pro.py switch to semantic fallback generators that tailor outlines and stories based on user parameters, preventing 500 errors.
- Concurrent Inking with Cache Check: In image_generator.py, prompt strings are hashed to check for existing panel assets before initiating redundant generation.
- Unicode Font Handling: TrueType fonts are loaded with error-handling to prevent character set encoding exceptions in multi-language environments.